



DEDAN KIMATHI UNIVERSITY OF TECHNOLOGY 2025/2026
UNIVERSITY EXAMINATION 2025/2026
THIRD YEAR, FIRST SEMESTER
BACHELOR OF SCIENCE IN MECHATRONIC ENGINEERING

EMT 3105: CIRCUIT AND NETWORK THEORY II

TIME: 11-1PM

DATE: 12TH AUGUST 2025

INSTRUCTIONS

1. This examination contains FIVE questions.
2. Question 1 is Compulsory and carries 30 Marks. All the rest carry 20 Marks each
3. Answer ONE Question from Section A and ONE Question from Section B
4. Common Unilateral Laplace Transforms Table is provided on the last page

QUESTION ONE (30 MARKS)

- a) State two differences between lag network and lead network (2 marks)
- b) Solve the electric circuit in Figure Q1(b) by using Laplace transformation for final steady-state current (5 marks)

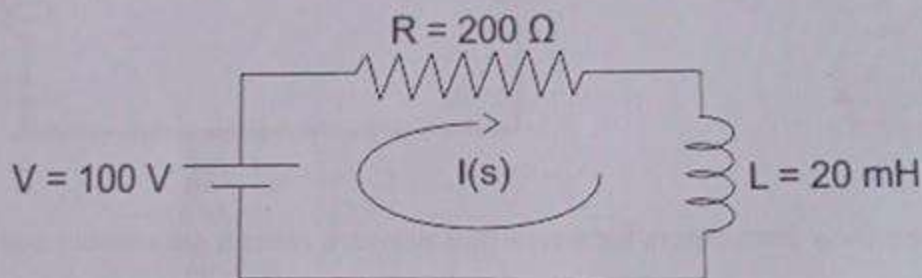


Figure Q1 (b)

- c) For the transfer function given, sketch the bode diagram which shows how the phase and magnitude of the system are affected by changing input frequency (5 marks)

$$TF = \frac{100000(s + 1)}{(s + 10)(s + 1000)}$$

d) Find Laplace inverse transform of

$$\left\{ \frac{s+1}{s(s+2)} \right\}$$

(5 marks)

e) Determine the y parameters of the network in Figure Q1 (e)

(5 marks)

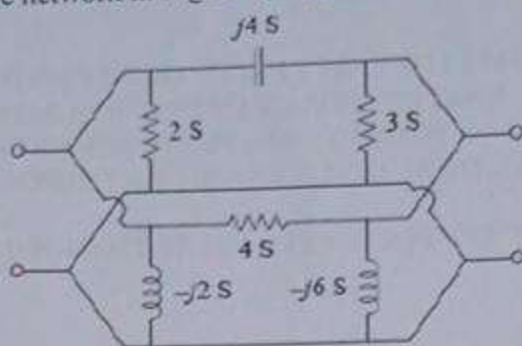


Figure Q1 (e)

f) Given $v(t) = 40u(t)$ V in Fig Q1 (f) find the current and voltage across the inductor

(5 marks)

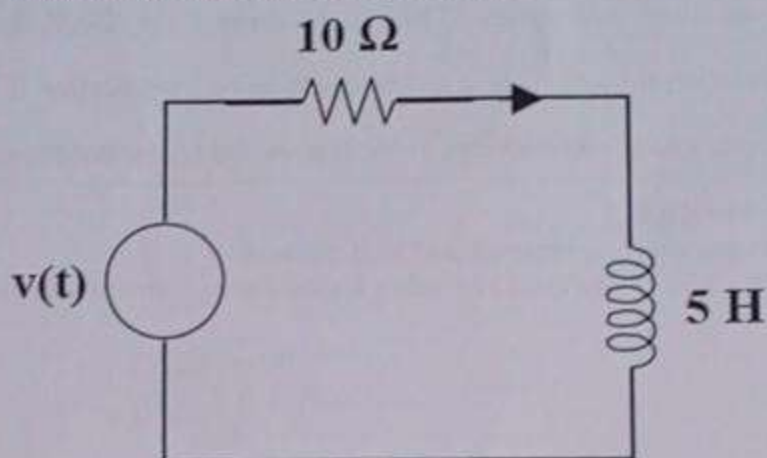


Figure Q1 (f)

g) Determine the y parameters for a two-port network if the h parameters are:

(3 marks)

$$h = \begin{bmatrix} 5 & 2 \\ 3 & 6 \end{bmatrix}$$

SECTION A

QUESTION TWO (20 MARKS)

Design 5th order unity gain Butterworth filter with the corner frequency $f_c = 50$ kHz using Table 5 (20 marks)

Table 5. Butterworth Coefficients

n	i	a_i	b_i	$k_i = \frac{f_{ci}}{f_c}$	Q_i
1	1	1.0000	0.0000	1.000	—
2	1	1.4142	1.0000	1.000	0.71
3	1	1.0000	0.0000	1.000	—
	2	1.0000	1.0000	1.272	1.00
4	1	1.8478	1.0000	0.719	0.54
	2	0.7654	1.0000	1.390	1.31
5	1	1.0000	0.0000	1.000	—
	2	1.6180	1.0000	0.859	0.62
	3	0.6180	1.0000	1.448	1.62
6	1	1.9319	1.0000	0.676	0.52
	2	1.4142	1.0000	1.000	0.71
	3	0.5176	1.0000	1.479	1.93
7	1	1.0000	0.0000	1.000	—
	2	1.8019	1.0000	0.745	0.55
	3	1.2470	1.0000	1.117	0.80
	4	0.4450	1.0000	1.499	2.25
8	1	1.9616	1.0000	0.661	0.51
	2	1.6629	1.0000	0.829	0.60
	3	1.1111	1.0000	1.206	0.90
	4	0.3902	1.0000	1.512	2.56

QUESTION THREE (20 MARKS)

Find $v(t)$, $i(t)$ for $t > 0$ consider three cases in Fig Q3:

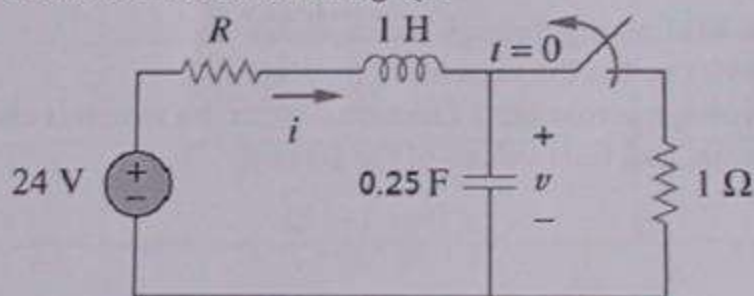


Figure Q3

- Case one $R = 5 \Omega$
- Case two $R = 4 \Omega$
- Case three $R = 1 \Omega$

(6 marks)

(6 marks)

(8 marks)

SECTION B

QUESTION FOUR (20 MARKS)

(10 Marks)

- a) Determine the admittance parameters of the T network in the Fig 4Q (a).

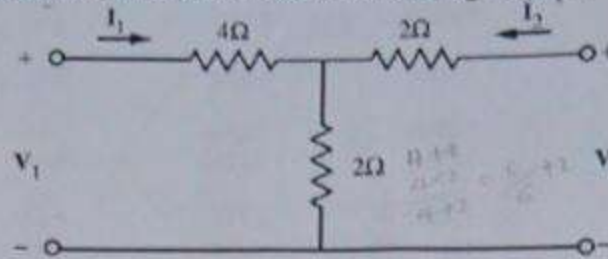


Figure Q4 (a)

- b) Find the step responses $v(t)$ and $i(t)$ to $v_s = 5u(t)$ V in the circuit of Fig Q4 (b)

(10 Marks)

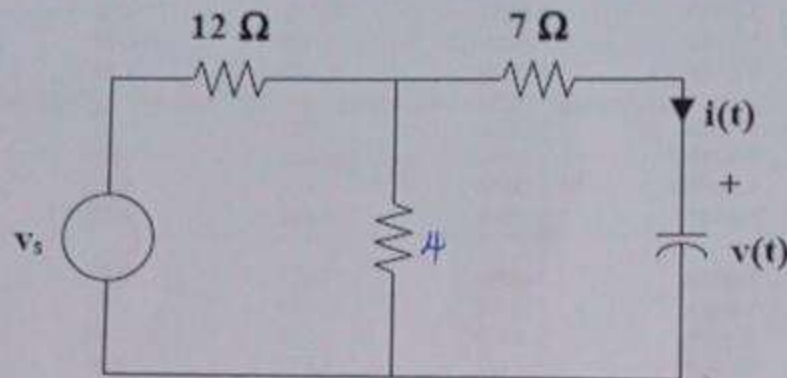


Figure Q4 (b)

QUESTION FIVE (20 MARKS)

In the circuit of Fig Q5

- Find the currents $i(t)$ flowing through the ~~capacitor~~ ^{inductor}
- Find the voltage $v(t)$ across the ~~capacitor~~ ^{inductor}
- Find the output voltage across the 5Ω resistor when the switch is closed
- Determine the initial and final values of the current

(4 marks)

(4 marks)

(4 marks)

(8 marks)

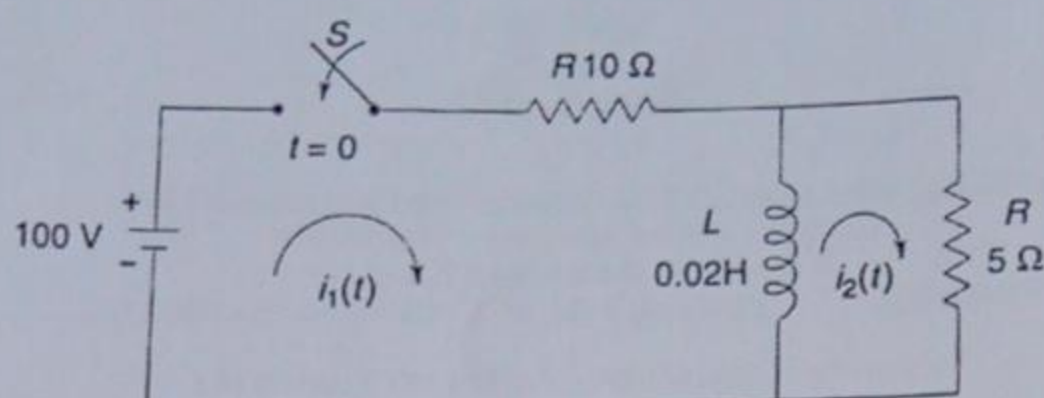


Figure Q5

Table 1 Common Unilateral Laplace Transform Pairs

$x(t)$	$X(s)$	ROC
$\delta(t)$	1	Entire s -plane
$u(t)$	$\frac{1}{s}$	$\text{Re}\{s\} > 0$
t	$\frac{1}{s^2}$	$\text{Re}\{s\} > 0$
e^{-at}	$\frac{1}{s+a}$	$\text{Re}\{s\} > -a$
$\sin(\omega t)$	$\frac{\omega}{s^2 + \omega^2}$	$\text{Re}\{s\} > 0$
$\cos(\omega t)$	$\frac{s}{s^2 + \omega^2}$	$\text{Re}\{s\} > 0$
te^{-at}	$\frac{1}{(s+a)^2}$	$\text{Re}\{s\} > -a$
$e^{-at} \sin(\omega t)$	$\frac{\omega}{(s+a)^2 + \omega^2}$	$\text{Re}\{s\} > -a$
$e^{-at} \cos(\omega t)$	$\frac{s+a}{(s+a)^2 + \omega^2}$	$\text{Re}\{s\} > -a$